

## BALICE, Poland

WMO: 125660

Lat: 50.0777N

Lon: 19.7948E

Elev: 241

StdP: 98.46

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -15.6      | -12.5 | -18.0                         | 0.8 | -15.6 | -14.8 | 1.1 | -12.1 | 12.3                  | 5.7  | 11.0 | 4.7  | 2.2                   | 60   | 0.537 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.7                   | 30.4            | 20.5 | 28.5 | 19.8 | 26.8 | 19.0 | 21.5                | 28.2 | 20.7 | 26.9 | 19.8 | 25.2 | 3.1                  | 90   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 19.2                            | 14.4 | 24.1 | 18.5 | 13.8 | 23.2 | 17.8 | 13.2 | 22.3 | 63.7          | 28.4 | 60.6 | 27.1 | 57.5 | 25.3 | 25.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 9.7           | 8.4  | 7.3 | -18.2                      | 33.6 | 4.0                | 1.5 | -21.1                                              | 34.7 | -23.4      | 35.5 | -25.6      | 36.4 | -28.5      | 37.5 |    |    |
| (5)               |      |     | -18.5                      | 23.5 | 3.9                | 0.8 | -21.3                                              | 24.0 | -23.5      | 24.4 | -25.7      | 24.9 | -28.5      | 25.4 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 9.0    | -1.9  | -0.4  | 3.5   | 9.3   | 14.1  | 17.5  | 19.4  | 18.8  | 14.0  | 9.1   | 4.2   | -0.6  |  |  |
|                                                                        | DBStd       | 8.58   | 5.32  | 5.03  | 4.32  | 4.15  | 3.67  | 3.41  | 3.09  | 3.20  | 3.39  | 4.10  | 4.16  | 4.93  |  |  |
|                                                                        | HDD10.0     | 1511   | 370   | 291   | 203   | 61    | 7     | 0     | 0     | 0     | 6     | 69    | 177   | 328   |  |  |
|                                                                        | HDD18.3     | 3569   | 628   | 524   | 459   | 273   | 139   | 56    | 26    | 33    | 134   | 288   | 423   | 587   |  |  |
|                                                                        | CDD10.0     | 1131   | 0     | 0     | 2     | 39    | 133   | 224   | 290   | 274   | 126   | 39    | 4     | 0     |  |  |
|                                                                        | CDD18.3     | 147    | 0     | 0     | 0     | 1     | 7     | 30    | 57    | 48    | 4     | 0     | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 1477   | 0     | 0     | 0     | 18    | 99    | 312   | 543   | 442   | 61    | 2     | 0     | 0     |  |  |
|                                                                        | CDH26.7     | 408    | 0     | 0     | 0     | 2     | 15    | 82    | 169   | 131   | 9     | 0     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 2.9    | 3.3   | 3.3   | 3.5   | 3.1   | 2.7   | 2.8   | 2.7   | 2.3   | 2.6   | 2.6   | 2.9   | 3.0   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 696    | 35    | 31    | 34    | 50    | 80    | 98    | 94    | 83    | 55    | 47    | 43    | 39    |  |  |
|                                                                        | PrecMax     | 962    | 81    | 110   | 80    | 101   | 157   | 197   | 285   | 186   | 108   | 160   | 91    | 86    |  |  |
|                                                                        | PrecMin     | 470    | 6     | 1     | 2     | 13    | 34    | 41    | 40    | 24    | 6     | 1     | 8     | 5     |  |  |
|                                                                        | PrecStd     | 118    | 17    | 19    | 20    | 25    | 31    | 37    | 57    | 38    | 26    | 33    | 18    | 19    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 11.9  | 13.4  | 18.9  | 25.8  | 28.6  | 31.9  | 33.0  | 33.0  | 27.9  | 23.0  | 17.8  | 12.2  |  |  |
|                                                                        |             | MCWB   | 8.3   | 8.5   | 10.8  | 14.9  | 18.0  | 20.8  | 21.1  | 21.2  | 18.6  | 16.2  | 12.6  | 8.5   |  |  |
|                                                                        | 2%          | DB     | 8.5   | 10.3  | 15.7  | 22.4  | 25.9  | 29.1  | 30.6  | 30.0  | 25.0  | 20.4  | 14.8  | 9.2   |  |  |
|                                                                        |             | MCWB   | 6.5   | 7.0   | 9.5   | 13.7  | 17.3  | 20.2  | 20.5  | 20.6  | 17.7  | 15.0  | 11.1  | 7.1   |  |  |
|                                                                        | 5%          | DB     | 6.5   | 8.2   | 13.0  | 19.8  | 23.9  | 26.8  | 28.4  | 27.8  | 22.8  | 18.0  | 12.2  | 7.3   |  |  |
|                                                                        |             | MCWB   | 5.0   | 5.6   | 8.0   | 12.4  | 16.6  | 19.1  | 19.8  | 19.9  | 16.7  | 13.9  | 9.8   | 5.7   |  |  |
|                                                                        | 10%         | DB     | 4.8   | 6.2   | 10.5  | 17.2  | 21.9  | 24.8  | 26.3  | 25.7  | 20.6  | 15.6  | 10.2  | 5.8   |  |  |
|                                                                        |             | MCWB   | 3.6   | 4.3   | 7.1   | 11.1  | 15.3  | 18.0  | 19.0  | 18.9  | 15.8  | 12.9  | 8.7   | 4.6   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 8.7   | 9.1   | 11.5  | 15.9  | 19.6  | 22.4  | 22.7  | 22.6  | 19.7  | 17.0  | 13.0  | 9.3   |  |  |
|                                                                        |             | MCDB   | 11.0  | 12.3  | 16.9  | 23.6  | 26.1  | 29.3  | 29.9  | 28.9  | 25.3  | 21.8  | 16.2  | 11.4  |  |  |
|                                                                        | 2%          | WB     | 6.6   | 7.3   | 10.0  | 14.3  | 18.1  | 21.0  | 21.5  | 21.3  | 18.4  | 15.5  | 11.5  | 7.2   |  |  |
|                                                                        |             | MCDB   | 8.2   | 9.9   | 14.7  | 21.0  | 24.0  | 27.3  | 28.2  | 28.1  | 23.2  | 19.2  | 14.4  | 8.9   |  |  |
|                                                                        | 5%          | WB     | 5.0   | 5.8   | 8.7   | 13.1  | 17.0  | 19.9  | 20.6  | 20.5  | 17.3  | 14.3  | 10.1  | 5.9   |  |  |
|                                                                        |             | MCDB   | 6.3   | 7.9   | 12.4  | 18.6  | 22.8  | 25.5  | 26.7  | 26.6  | 21.7  | 17.5  | 11.9  | 7.2   |  |  |
|                                                                        | 10%         | WB     | 3.5   | 4.4   | 7.3   | 11.7  | 15.9  | 18.6  | 19.7  | 19.5  | 16.2  | 13.0  | 8.7   | 4.6   |  |  |
|                                                                        |             | MCDB   | 4.6   | 6.1   | 10.2  | 16.3  | 20.8  | 23.5  | 24.9  | 24.6  | 19.9  | 15.5  | 10.2  | 5.5   |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 6.0   | 7.0   | 8.9   | 10.8  | 10.8  | 10.4  | 10.7  | 10.9  | 9.8   | 9.1   | 6.9   | 5.9   |  |  |
|                                                                        |             | MCDBR  | 8.4   | 10.5  | 14.3  | 15.9  | 14.8  | 14.2  | 14.6  | 14.5  | 14.1  | 13.4  | 11.2  | 8.1   |  |  |
|                                                                        |             | MCWBR  | 6.5   | 7.4   | 8.7   | 8.2   | 7.3   | 6.6   | 6.2   | 6.4   | 7.5   | 8.3   | 7.9   | 6.3   |  |  |
|                                                                        |             | MCDBR  | 8.0   | 9.8   | 12.9  | 14.3  | 13.5  | 13.0  | 13.0  | 13.1  | 12.5  | 12.4  | 10.0  | 7.7   |  |  |
|                                                                        | 5% WB       | MCWBR  | 6.4   | 7.2   | 8.3   | 7.7   | 7.1   | 6.4   | 5.9   | 6.1   | 7.0   | 7.9   | 7.4   | 6.1   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.320 | 0.347 | 0.393 | 0.449 | 0.442 | 0.444 | 0.461 | 0.453 | 0.418 | 0.380 | 0.350 | 0.315 |  |  |
|                                                                        | taud        |        | 2.261 | 2.228 | 2.176 | 2.082 | 2.153 | 2.188 | 2.153 | 2.177 | 2.235 | 2.312 | 2.333 | 2.323 |  |  |
|                                                                        | Ebn at Noon |        | 697   | 773   | 798   | 791   | 818   | 818   | 796   | 780   | 765   | 727   | 658   | 651   |  |  |
|                                                                        | Edh at Noon |        | 72    | 96    | 121   | 148   | 144   | 140   | 143   | 133   | 113   | 88    | 68    | 60    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 0.85  | 1.59  | 2.71  | 4.04  | 4.91  | 5.22  | 5.02  | 4.54  | 3.22  | 1.94  | 0.96  | 0.66  |  |  |
|                                                                        | RadStd      |        | 0.15  | 0.26  | 0.40  | 0.49  | 0.61  | 0.51  | 0.61  | 0.38  | 0.47  | 0.28  | 0.13  | 0.08  |  |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | N/A   | N/A     | N/A    | +1.07   | +0.38 | N/A   | N/A         | N/A     | N/A     | +32     |
| (47) Regional (39 neighbors) |  | +0.55 | N/A     | N/A    | +0.88   | +0.45 | +0.38 | -128        | -179    | +80     | +33     |

Nomenclature: See separate page